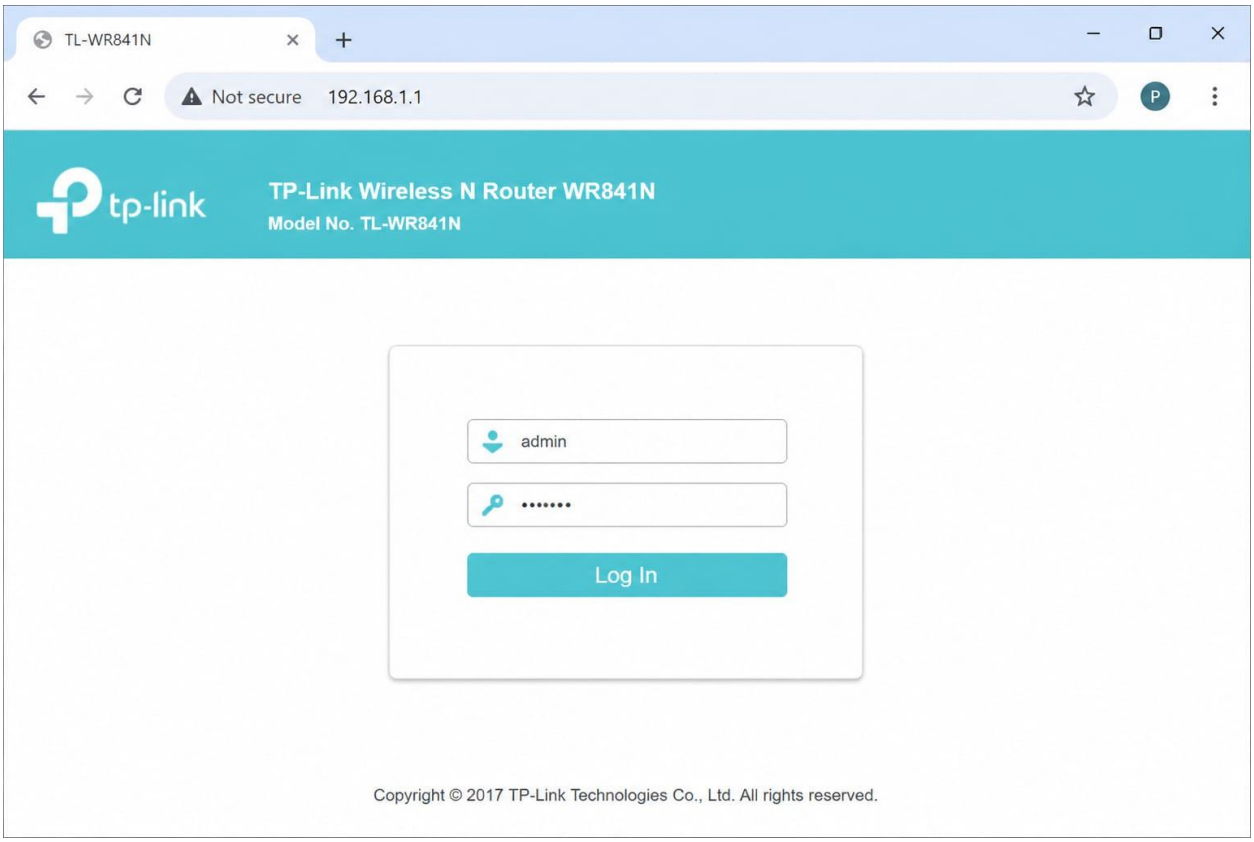
Q1. Log in to the Router's Admin Panel



1. Connect your laptop or mobile phone to the Wi-Fi network.

2. Open a web browser.

3. Enter the router's IP address, such as 192.168.1.1 or 192.168.0.1.

4. Enter the admin username and password.

5. The router's admin panel will open.

Q2. Find the IP Address, Subnet Mask, and Default Gateway

Answer:

The IP address identifies my device on the network. The subnet mask identifies the network portion of the IP address, and the default gateway is the router address used to access other networks.

Example values:

Network Setting	Example
IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

Q3. Change the Device's IP Address from DHCP to Static IP

Answer:

1. Open the network settings on your device.

2. Select your connected Wi-Fi network.

3. Find IP settings and change DHCP/Automatic to Static.

4. Enter the following example settings, only if they match your network:

   IP Address: 192.168.1.50

   Subnet Mask: 255.255.255.0

   Gateway: 192.168.1.1

   DNS: 8.8.8.8

5. Save the settings and test the internet connection.

6. After testing, change the setting back to DHCP/Automatic.

Result: A static IP address can work when configured correctly, but it must not conflict with another device's IP address.

Q4. Find the DHCP Range and Explain Duplicate IP Addresses

Answer:

The DHCP range is the range of IP addresses that a router automatically assigns to connected devices.

Example DHCP range: 192.168.1.100 to 192.168.1.200

If two devices are assigned the same IP address, an IP conflict occurs. This can cause connection problems, interrupted internet access, or difficulty communicating with other devices.